



SIP STUDIES

Business Plan with AI Integration

ABSTRACT

Think about taking the luxury of wine as a means for helping people in need with water accessibility. Now integrate it with the capabilities of artificial intelligence. Tie it all together with education and entertainment. Together, you have Sip Studies.

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Certification



MASSACHUSETTS INSTITUTE OF TECHNOLOGY
SLOAN SCHOOL OF MANAGEMENT

THIS IS TO CERTIFY THAT

Jonathan Yu

HAS SUCCESSFULLY COMPLETED THE EXECUTIVE PROGRAM

Artificial Intelligence: Implications for Business Strategy

August 2023

A handwritten signature in black ink, reading "Peter Hirst". The signature is written in a cursive style with a horizontal line underneath.

PETER HIRST
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0359167

Introduction

Sip Studies is a beverage education and entertainment organization focused on beverage accessibility, from water to wine, with differentiating high-quality products, services and charities that will utilize artificial intelligence. It is this that we must ask: what is artificial intelligence?

- Breaking the term down to each word's core, according to MIT, "**artificial** is more straight forward, meaning: something that does not occur naturally." They continue with a definition by psychologist Howard Gardner: "**Intelligence** is the ability to solve problems, or to create products, that are valued within one or more cultural settings." (Gardner 1983)
- Professor Patrick Winston then follows the concept with the compounding reasons for AI's meaning: "AI is about the architectures that deploy methods, enabled by constraints, exposed by representations, that support models of thinking, perception, and action."
- During my time taking MIT's course on "Artificial Intelligence: Implications for Business Strategy", I began to formulate my own definition for intelligence applicable for machines and nature alike, guided by the writings of Sir Kenneth Robinson:
 - "We are all born with extraordinary powers of imagination, intelligence, feeling, intuition, spirituality, and of physical and sensory awareness..." (*Robinson 9*)
 - "Physiologists largely agree that in addition to the five [senses] we all know about, there are four more." (*Robinson 32*)
 - **Temperature** (Thermoception)
 - **Pain** (Nociception)
 - **Vestibular** (Equilibrioception)
 - **Kinesthetic** (Proprioception)
 - With **Synesthesia** as a feature
 - As a result, I would unpack **Intelligence** as the following:
 - A collection of learning sensors capable of gathering, transferring, and unpacking vast amounts of data that have been filtered by various tools of specialization.
 - It is the speed, cache, and efficiency in which an intelligence is measured, which then is held accountable through real-life backpropagation, called "practice" or "experimentation".
 - It is through our senses that different forms of intelligence arise.
 - "Application" of your memories and skills are the results of intelligence.

With this perception of intelligence in mind, we will build a multi-faceted organization where humans and machines will build a complex web of intelligence together. There are five overall aspects of artificial intelligence technology that we will incorporate into our training, operations, and products:

- **(ML)** Machine Learning
 - "ML as a discipline, tries to design, understand, and use computer programs that learn from experience (i.e. data) for the purpose of modeling, prediction and control." (MIT)

- ML can be used in two main ways, sensing and predicting.
 - *Sensing*: processing large amounts of data; image recognition, & image analysis.
 - *Predicting*: using data to make future assessments, personalizing product offerings, & improving product performance.
- **(NLP)** Natural Language Processing
 - NLP “is primarily concerned with giving computers the ability to support and manipulate speech... The technology can then accurately extract information and insights contained in the documents as well as categorize and organize the documents themselves.” (Wikipedia)
 - NLP will be utilized for search, information extraction, machine translation, spam detection, text and voice generation, question answering, as well as sentiment analysis.
- **(RPA)** Robotic Process Automation & Robotics
 - “A robot is an autonomous machine capable of sensing its environment, carrying out computations to make decisions, and performing actions in the real world.” (RobotsGuide.com)
 - There are two facets of robotics that will be addressed, robots and RPA.
 - *Robots*: Working with NLP and LLM to Sense, Compute, and Act in the real world.
 - *RPA*: Working with ML for customer services, website support, educational guidance, staff training, and teaching support.
- **(LLM)** Large Language Model
 - “Large language models (LLMs) are deep learning algorithms that can recognize, summarize, translate, predict, and generate content using very large datasets.” (Nvidia)
 - The premise of building an LLM will be to build our own, in-house model with beverage data from suppliers, distributors, restaurants, retailers, customers and more. This will primarily serve as an archive for historical data, industry and market knowledge, as well as preserve technological advances. It will interplay with all our AI facets of ML, NLP, and Robotics.

Executive Summary

Think about taking the luxury of wine to help people in need with water access. My start-up company is a beverage education and entertainment organization designed to leverage the use of artificial intelligence in order to test the limits of human capability and compassion, one employee at a time, in order to help those in need of water accessibility. AI establishes a foundational base of **Differentiating Business Strategies** by supporting the **high-quality services, products, and charities** we intend to produce – leading the beverage category with a stronger web of knowledge. Our vision is to:

- Successfully educate while entertaining people about beverages, from water to wine.
- Help people drink better, by gaining access to cleaner water and artisanal flavors.
- Connect, support, and archive knowledge and data from all contributing companies to analyze, grow, and protect beverage knowledge collectively.
- Build beverage technology that amplifies the human effort to Learn, Engage, and Teach about beverages.

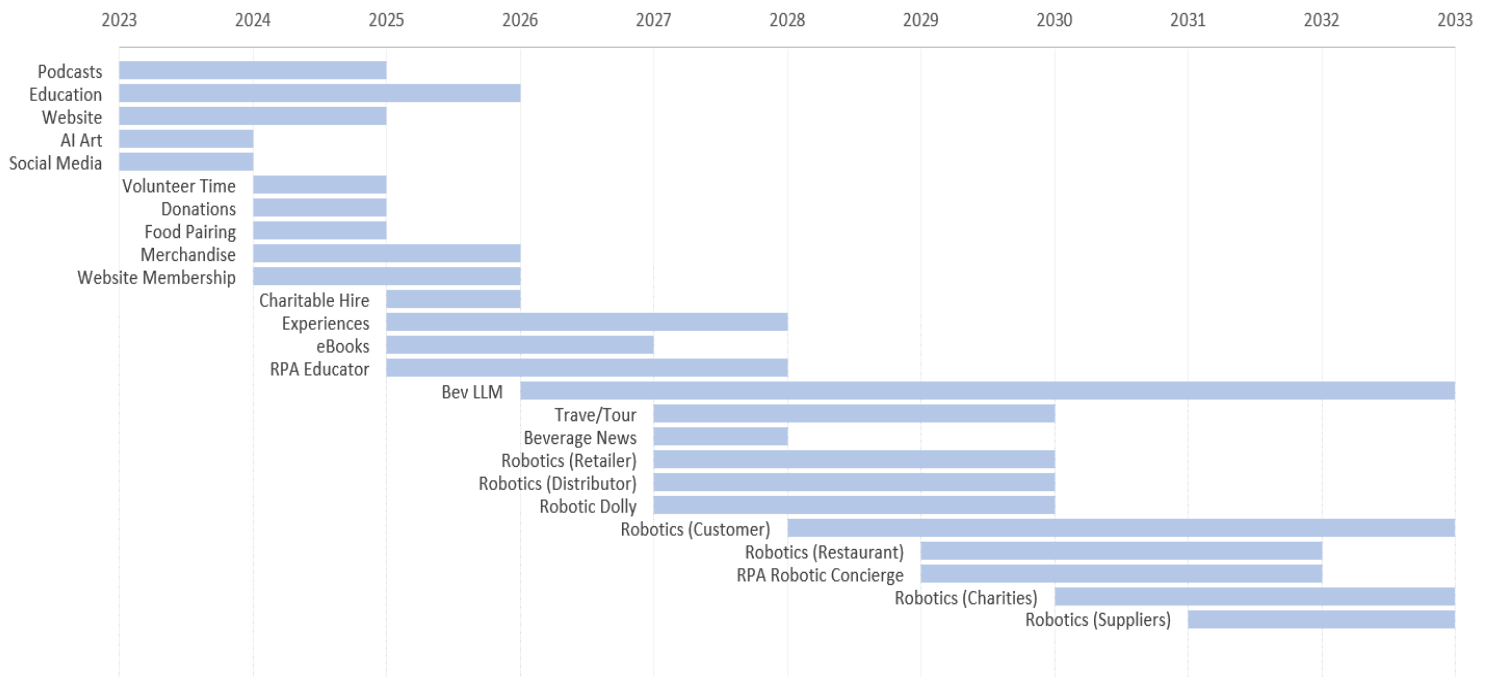
Implementing AI creates a doppelganger for each employee with a base goal of 50% increased productivity and capability based off Study 2 of a Neilsen Norman Group study (*Neilsen*), compensating for productivity gaps with:

- AI being an amplifying teammate and teacher to our human capabilities, and
- Humanity being a vigilant guide and designer to our technological capabilities.

The strategy for implementing AI will begin with:

- **Phase One Initial Growth (1-3 years):** Employees using 3rd person party services to build our foundation quickly, scaling our services, products, and charities.
- **Phase Two Integrations (3-5 years):** Internal development of ML, NLP, RPA, & LLM data collection to establish AI services and products in-house that will extract data and enhance experiences to raise our customer value. This will also be our first phase of upskilling all employee skillsets and knowledge.
- **Phase Three Integrations (5-8 years):** Manufacturing robotic beverage technology for suppliers, distributors, retailers, customers, and charities. We will be utilizing our data to build solutions for these external stakeholders with accuracy and variety. This will be our second phase of major upskilling all employee skillsets and knowledge.

The diagrammatic table below is for the overall products, services and charities, most we intend to integrate with AI:



Over the span of the first ten years, the ultimate ikigai goal for the company is to assist and create water accessibility solutions through the education of all beverages. It will be a culmination of one-to-three-year projects that will establish the products, services and charitable causes we desire to create.

Phase One will focus on utilizing and understanding AI systems currently for production and education as each member weaves its use into the day-to-day operations.

Phase Two will allow us to bring the AI technology in-house, with significant reinvestment in our skillsets, capabilities, and enhance our customer values and extract data to elevate these features.

Phase Three is our company's efforts on integrating mobile RPA solutions with robotics to bring beverage solutions to our customers directly.

To strengthen, expand, and explore our beverage prowess, we will need the differentiation of AI to increase the collective intelligence of both employees and technology.

Current State

The current state of technology in my company has been rudimentary with:

- Squarespace web-hosting, data extraction, and design
- Graphic design and video editing
- Live streaming and recording for media uploads, and
- Implementing 3rd party AI services to our

Phase 1 Year 1 Products below:

Task	Product	Phase	Cost	Category	AI	Integration Type	Department	AI Output Strategy
1	Podcasts	1 - Initial Growth	Free	Service	Yes	Enhance	CRM	ML Content Creation
2	Education	1 - Initial Growth	Economy	Service	Yes	Both	CRM	RPA
3	Website	1 - Initial Growth	Professor	Service	Yes	Both	CRM	ML, NLP, RPA Guide
4	AI Art	1 - Initial Growth	Economy	Product	Yes	Enhance	eComm	ML Content Creation
5	Social Media	1 - Initial Growth	Free	Service	Yes	Both	eComm	ML Content Creation

Currently the company has used some generative AI to rebuild the company business plan, market strategy, and branding.

The philosophical pillars of “Learn, Engage, Teach” are an example of how incorporating innovation is part of our dynamic foundation. This has allowed for recalibration when approaching the competitive market, rebuilding products and services for all of our customers, and identifying charitable causes that will drive our social goals in the communities most in need.

Our current beverage competition resides in a few categories:



Guilds	Websites	eComm	Maps	Journalism	Books	Apps
SWE	Youtube	TotalWine.com	Wine Foley	Wine Spectator	Oxford Compendium	Khan Academy
CMS	Guild Somm	liquor.com	De Long	Wine Enthusiast	Wine Bible	Coursera
WSET	Wine Foley	drizly.com	SWET	Wine Advocat	Windows to the World	Duolingo
ISG	Wine Wit & Wisdom	untappd.com	CMS	BeerNet	World Atlas of Wine	Vivino
WSG	Seven Fifty	wine-searcher	Burgundy Maps	CraftBusinessDaily	World of Wine	UnTapped
ABA	Neilson IQ	wine.com	Regional Areas	WineSpiritsDaily	Postmodern Winemaking	TheSpirits Business

These competitors are mostly in the Adult Beverage category as there are the most legal restrictions involved with this category from shipping, distributing, merchandising, and more. In the United States, the Alcohol and Tobacco **Tax and Trade Bureau (TTB)**, under the Department of Treasury, is the federal regulator and tax collector on trade and import products. In other nations, they are typically regulated with the same department as food regulation instead of tobacco and firearms. We will have most of our obstacles in this category and can utilize AI to navigate these unknown obstacles.

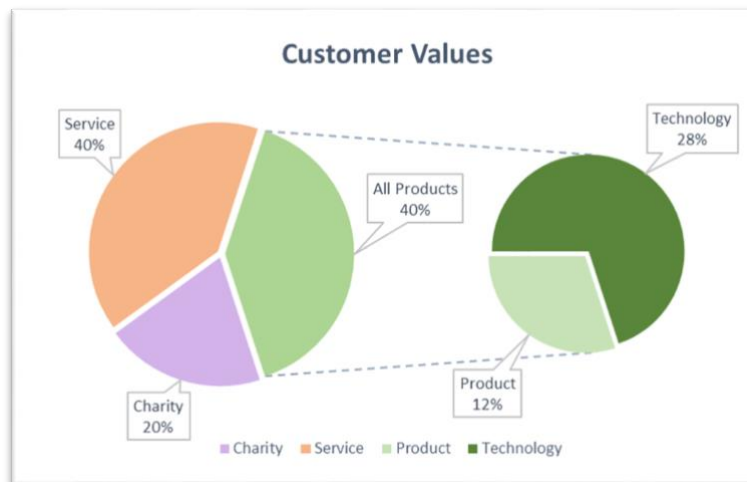
By uniting beverages as a whole category, particularly with adult beverages initially, it will be the overarching collective intelligence of all our companies and customers with which to **connect, support, and archive beverage knowledge**. This would be like a digital Library of Alexandria. The goal would be to build an LLM to archive and protect the ever-changing knowledge, skills, and technology of beverage consumption from catastrophic cultural changes like US Prohibition in 1920, which destroyed vast amounts of technology and knowledge setting the industry back 50 years.

By building an LLM, we will be establishing a tremendous differentiating foundation in which customers would be able to interact with the data using an online RPA through NLP in ways that

allow our students to interact, learn, and be mentored by the knowledge extracted from the industry data, statistics, and trends.

This will be a critical part of our second phase which will also tie into our goals in ML generative AI, not only for creating a teaching aid for our customers but also for our employees to learn from too. This is the gateway for our training data to actually guide our employees and leverage all possibilities to learn faster, research from a vast array of concentrations, and propel our products, services and charities.

Serving as a foundation, this model will also be the basis for our Robotics solutions as we use it to bring accessibility closer to all our customers with NLP interactions, wayfinding, data extraction and sentiment analysis for a majority of our products.



Proposed Initiative

We will be covering the whole technology gambit of the services, products, charities and technology planned for each phase of this start-up company. The proposed strategies for our products are the following:

Task	Product	Phase	Cost	Category	AI	Integration Type	Department	AI Output Strategy
1	Podcasts	1 - Initial Growth	Free	Service	Yes	Enhance	CRM	ML Content Creation
2	Education	1 - Initial Growth	Economy	Service	Yes	Both	CRM	RPA
3	Website	1 - Initial Growth	Professional	Service	Yes	Both	CRM	ML, NLP, RPA Guide
4	AI Art	1 - Initial Growth	Economy	Product	Yes	Enhance	eComm	ML Content Creation
5	Social Media	1 - Initial Growth	Free	Service	Yes	Both	eComm	ML Content Creation
6	Volunteer Time	1 - Initial Growth	Free	Charity	No	N/A	Everyone	N/A
7	Donations	1 - Initial Growth	Economy	Charity	No	N/A	Everyone	N/A
8	Food Pairing	1 - Initial Growth	Health	Service	Yes	Extract	CRM	Collect MDM Data
9	Merchandise	1 - Initial Growth	Health	Product	Yes	Enhance	eComm	ML Content Creation
10	Website Membership	1 - Initial Growth	Professional	Service	Yes	Both	CRM	ML, NLP, RPA Guide
11	Charitable Hire	1 - Initial Growth	Health	Charity	Yes	Enhance	HRM	ML Filtration
12	Experiences	1 - Initial Growth	Luxury	Charity	Yes	Enhance	CRM	NLP Travel Planning
13	eBooks	1 - Initial Growth	Free	Product	Yes	Both	eComm	ML Content Creation
14	RPA Educator	2 - First Phase	Premium	Service	Yes	Both	IT	ML, NLP, RPA Guide
15	Bev LLM	2 - First Phase	Luxury	Service	Yes	Extract	IT	Collect MDM Data
16	Travel/Tour	2 - First Phase	Luxury	Service	Yes	Both	CRM	NLP Travel Planning
17	Beverage News	2 - First Phase	Free	Service	Yes	Enhance	eComm	ML, NLP, RPA Guide
18	Robotics (Retailer)	3 - Second Phase	Luxury	Technology	Yes	Both	MRP/Manuf.	Robotics labor help
19	Robotics (Distributor)	3 - Second Phase	Luxury	Technology	Yes	Both	MRP/Manuf.	Robotics labor help
20	Robotic Dolly	3 - Second Phase	Luxury	Technology	Yes	Enhance	MRP/Manuf.	Robotics labor help
21	Robotics (Customer)	3 - Second Phase	Luxury	Technology	Yes	Both	MRP/Manuf.	RPA Robotics Guide
22	Robotics (Restaurant)	3 - Second Phase	Luxury	Technology	Yes	Both	MRP/Manuf.	RPA Robotics Guide
23	RPA Robotic Concierge	3 - Second Phase	Luxury	Technology	Yes	Both	MRP/Manuf.	RPA Robotics Guide
24	Robotics (Charities)	3 - Second Phase	Luxury	Charity	Yes	Both	MRP/Manuf.	Robotic Water Accessibility
25	Robotics (Suppliers)	3 - Second Phase	Luxury	Technology	Yes	Both	MRP/Manuf.	Robotic Beverage Analysis

- This chart shows their implementation phase, their approximate cost category, and the type of category it is organized into (costs in order: Free, Economy, Health, Professional, Premium, then Luxury).
- The integration of AI in each product has also been identified for being an Enhancing feature for our clients, or an Extracting feature for our data pipelines. Should the products be both, these services will have user controls for consent and security in case customers should prefer not to upload their data.
- After phase two and three of hiring and upskilling, the products will be reorganized, maintained, and developed by their respective accountable departments, with much developmental crossover.

A detailed breakdown of the **AI Output Strategies** will identify the differentiating advantages of AI use in all of our products.

- **Collecting MDM Data** to support all AI facets with clean data, supervised learning and active learning types to build interpretable modeling for all AI outputs. We will keep humanity in the loop for guidance, organization, and ethical monitoring of information.
- **ML Content Creation** for scalable speed to market with design ideas on social media, websites, eBooks, and more to use sensing techniques like image recognition and image analysis to process and produce in-house content.
- **ML Filtration** used to support and identify ideal new hires faster to fill our growing company requirements

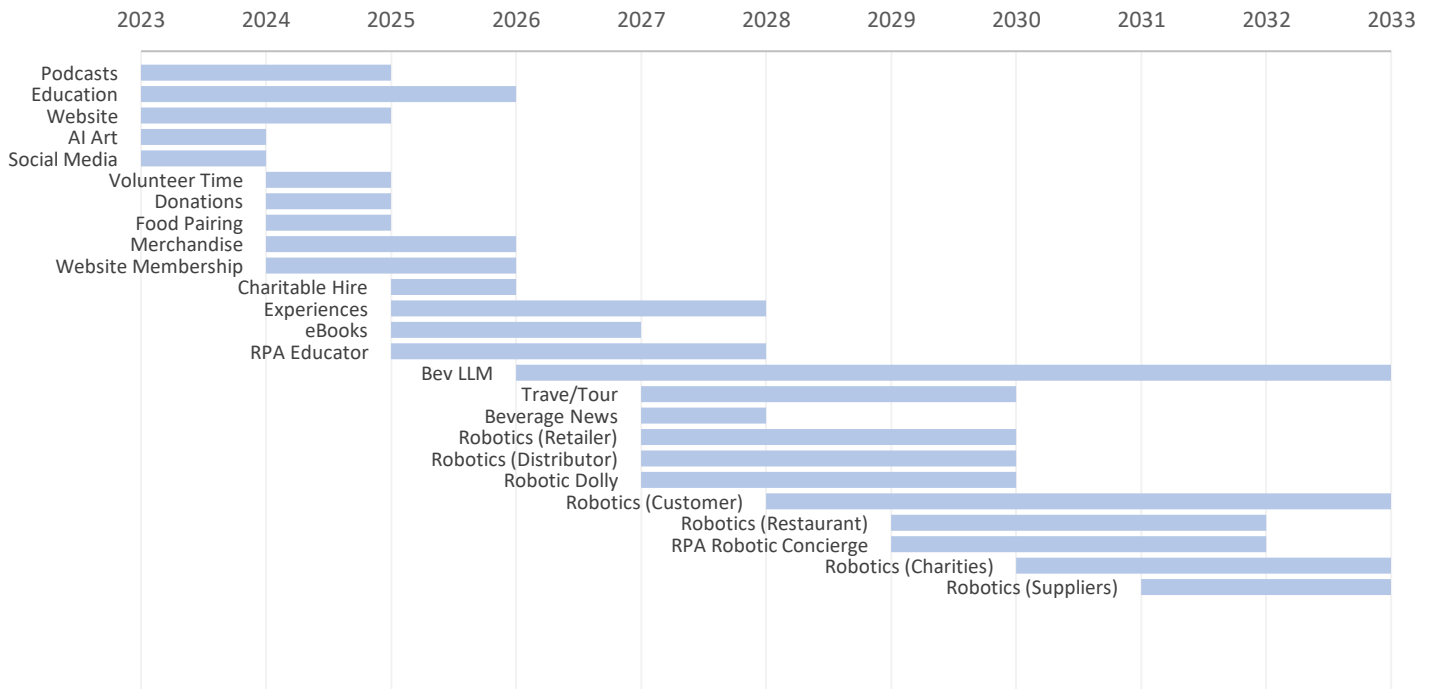
- **ML, NLP, RPA** will all be used in conjunction to establish an online beverage teacher to assist students in their path towards various beverage certification theory as well as a resource for teachers to expedite and elaborate answers for their students in real time.
- **NLP Travel Planning** will be a luxury product where we will be able to bring our clientele to beverage destinations around the world efficiently in all encompassing destination packages for learning.
- **Robotic Water Accessibility** will be an avenue for our MRP to design on site solutions to resolve complex site-by-site water accessibility challenges that can be resolved through engineering tactics. This will be the culmination of taking our resources to funnel charitable solutions around the world.
- **Robotics Labor Help** will be RPA Robotic Guides in retail and restaurants that will assist customers in way finding to tables and products, sentiment analysis for understanding the customer's needs around specific products, and recommendations base of on hand knowledge of current inventories. A Robotic Concierge.
- **Robotics Labor Help** will also have another avenue of RPA Robotic Dollies to assist suppliers, distributors, and retailers in repetitive labor. We will have MRP department build and integrate sensors and actuators to fine tune our robotics abilities to sense, compute and act in our customer's environments in scanning, lifting, and locating product locations for general warehouse assistance. The key here will eventually be building it to be more cost effective but our priority will be inventing and developing a safe and effective product.
- **Robotic Beverage Analysis** will be designed in a way to use ML image analysis for regional and global data analysis on all beverage problems. This will be a critical component for our robotics as we use it for navigation and wayfinding, object manipulation, and hazard avoidance.

It is through the technical requirements of establishing each proposed initiative where we need to be clear

- **Machine Learning** – We will utilize ML as our input sensors with supervised learning to maintain the embedded human-technological partnership in the company. Utilizing this for data analysis, accumulation of large data, image recognition, and personalization through better prediction.
- **Natural Language Processing** – Will initially start off as external services through generative AI resourcing, spam detection, and sentiment analysis. As we develop our internal models and capabilities, we need to dive deeper into complex problems like questions answering, paraphrasing, and deeper obstacles for fine tuning student-teacher connectivity.
- **Robotic Process Automation** – core designed for building, deploying, and managing our software robotics from educational teaching, increasing customer satisfaction, productivity and accuracy, scalability and speed to market, and maximizing profitability for future projects.
- **Robotics** – will be our last phase of implementation with the intent that our ML, NLP, and RPA services will provide more information, mapping, and compute power to jump start all the mechanical abilities. This will be the active torque wrench of our company, solving global problems as we encounter them, with intelligent and problem-solving support.

Plan of Action & Criteria for Success

The execution of each project will vary from one to three years, as a criterion of our growing and measurable success, with our proposed sequences of events to establish working, viable products in this roadmap included below:



In Phase 1, our skillset will need to be hired or developed as we build. Our focus will be on building educational foundations in the community, expanding our networking reach, and developing fundamental products for instant and quick regenerative revenue streams. The first departments that need to be established will be **CRM, Finance, and Human Resources**. Specific AI skillsets needed:

- Generative AI prompting
- Utilization on 3rd party genAI features within design, analytics, and information extraction.
- Elements that will benefit immediately for our ROI will be from content creation, merchandise, speed to market, and educational facilitation.

Department requirements will be -

Customer Resource Management:

- Beverage Educators
- Sales and Support
- Key Account Managers
- Customer Service

Financial Resource Management:

- Financial Manager
- Accounting

- Analyst
- Trade Development

Human Resource Management:

- Staff Coordinator
- Hiring Manager
- Staff Trainer

In Phase 2, we will elevate all employee skillsets in order to bring ML, NLP, LLM, and RPA in-house. This will be a huge undertaking for our company by building the next critical departments within the company: **IT and eCommerce**. The development of these four projects will be staggered as we build, rebuild, and masterplan our informational foundations, architecture, and network. Specific AI skillsets and positions needed will be:

- All staff learning Python, TensorFlow, PyTorch
- Mathematics and statistics in linear algebra, calculus, probability and more.
- Machine Learning fundamentals of supervised learning, regression and classification, neural networks, etc.
- NLP skills of search, information extraction, machine translation, text generation, sentiment analysis, and word-sense disambiguation.

Department requirements will be -

Information Technology Management:

- Security
- Network Admin
- Master Data Manager
- Systems Analyst and Architect
- Service and Support

eCommerce Management:

- Graphic Design
- Videographer
- Marketing
- Editing and Journalism

In Phase 3, we will raise the overall corporate skillsets by building, integrating, and restructuring our systems, networks, and operations with Robotics. The implementation of this phase will build the last required departments within the company: **Manufacture Resource Planning, Supply Chain Management, and Warehouse Resource Management**. With Specific AI skillsets and positions needed will be:

- Computation
- Sensor development
- Actuators and mechanics
- Communication

Department requirements will be -

Manufacture Resource Planning:

- Mechanical Engineering
- Electrical Engineering

- Environmental Engineering
- Chemists

Supply Chain Management:

- Analyst
- Demand Planner
- Inventory Manager
- Production Planner
- Procurement Manager & Purchasing

Warehouse Resource Management:

- Inventory Management
- Facilities Management

With each phase we will adjust our awareness and perception for our key stakeholders as our products will be built both for our growing external and internal stakeholders:

Stakeholders – External:

- Suppliers (From Bacardi to Coca Cola)
- Distributors (From Starbucks to Breakthru Beverage)
- Retailers
- Restaurants
- Customers (Actual end-consumers of the product)
- Charities

Stakeholders – Internal:

- Investors & Board of Directors
- Employees Phase 1
- Employees Phase 2
- Employees Phase 3
- Employee Families

As we build each phase, just like our AI training, our own development will require mapping, localization, computing, and acting with deep awareness of ground truth and any back-propagation to correct and adjust to our growth.

We will need to address some ethical obstacles as we develop and execute each project but as we grow, we will be able to develop an ethics team for biases. Some concerns:

Ethical Concerns

The danger of biased responses from online NLP RPA systems will be something that needs serious vetting as we ensure that the responses and answers are to the topic and don't cause harm to our students.

Other issues that will arise will be the danger of robotics in the workplace. They will need to be designed for human interaction so that safety remains the most important thing. These interacting robotic systems will need to take precautionary actions around customers, employees, and other equipment hazards.

Risk of Unemployment

Unemployment will always be a concern as we may jeopardize not only our own employees but also those of our clients. Here we need to design systems and robotics that involve, where possible, the human interaction as a guidance precaution and not a means of replacement.

Planning for Mistakes

Reviewing biases in training data that may need ethical review for issues of “denigration, stereotype, recognition, under-representation, and ex-nomination” (*Crawford, t22:18*) as well as review of real responses through a training phase before release. I appreciate Kate Crawford’s approach to “interdisciplinarity” with building a “FATE” group (Fairness, Accountability, Transparency, & Ethics) (*Crawford, t39:47*) as this provides a straight forward approach to start solving these problems.

For now, we will have to plan for initial narrow AI use cases to implement specific tasks that will be a more efficient way of mitigating ethical harm, unemployment, and physical dangers.

Roadmap Extension:

Any plans for roadmap extension will be determined by each project’s success and successor. Each project will interlay into the importance of the following project. So far, we have a lot ahead of us already, but for now it will best be said that any extension for our roadmap will aim at securing a more accessible future globally, for all people to have access to clean, sustainable, accessible water solutions is our priority. Once we can say we’ve moved the needle to global success, we will adjust our efforts to other global obstacles like carbon sequestration and agricultural solutions.

Citation:

- Crawford, K. (Speaker). (2017, Dec 10). *The Trouble with Bias – NIPS 2017 Keynote – Kate Crawford #NIPS2017* [Video]. YouTube.
https://www.youtube.com/watch?v=fMym_BKWQzk&t=772s
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